Saving and restoring a state

Writing the analysis to a file, and reading it back into a session.

AT A GLANCE

Where	Data → Restore
The file	A compressed .copylotstate file, named copylot_state_ and the date and time it was written.
Saves	Annotations and their metadata tables, filter and normalization settings, the imputation and its matrix, the condition palettes, the metadata profiles, which package and profile were active, and the state of the interface.
Does not save	The measured data. Datasets are recorded by filename and path and re-read from disk on restore.
Never writes	Your uploaded files, or the settings of the installation. A restore only reads them.
Two kinds of save	Ordinary, which records where the datasets are, and portable, which does not.

1 What a save state is

A save state records the analysis setup: decisions made about a dataset — how the samples were annotated, what was filtered, what was normalized, what was imputed, which colours the conditions have, which page was open — plus stored resources such as the realised imputed matrix. It can therefore be large and can contain scientific data. Original measurement files are still required on restore, either at their recorded locations or supplied as replacements. Keep them alongside the state.

2 Saving

Three buttons, under **Save State**:

Controls on this page

Control	What it does	Default	Where
Save State… / Update Save State	Writes the state. Once a save file has been chosen in this session, its name is shown above the buttons as the current save target and the button writes to it again.	—	page
Save State As…	Writes the state to a file chosen now, which then becomes the save target.	—	page
Save Portable State…	The same state with every local dataset path left out.	—	page

The difference between the last two matters. An ordinary state records the absolute path of each dataset file, so restoring it on the same machine finds the data without being asked. A portable state deliberately does not, so those local dataset paths are omitted — and every dataset has to be supplied by hand when it is restored. Save portable when the file is going to somebody else; save ordinary when it is for yourself tomorrow. Portable does not mean anonymous: filenames, labels, metadata and stored values may still identify the study. Review them before sharing.

3 Restoring

1. Select the file in the field under **Restore State**. Where the platform allows it, **Open Save File…** opens a proper file dialogue instead.
2. Press **Restore**.
3. If any dataset the file refers to cannot be found, nothing is changed: a **Missing Files** panel lists each one with its package, its level, the path that was recorded and the filename it had. Choose a replacement file for each, then press **Restore** again.
4. On success the page reports how many datasets and how many annotations were restored, and lists any warnings.

Controls on this page

Control	What it does	Default	Where
Save state file to restore	The .copylotstate file.	none until chosen	page

Controls on this page (continued)

Control	What it does	Default	Where
Open Save File...	A native file dialogue, where the platform provides one.	—	page
Restore	Runs the restore. Disabled until a file is chosen, and while a missing dataset has no replacement.	—	page
Replacement file	One per missing dataset, in the **Missing Files** panel.	none	page

What comes back

- The datasets, re-read from their recorded paths or from the replacement files supplied, for every package the state held.
- The metadata profiles and their tables, and which profile was active.
- For every annotation: the sample-to-condition table, the filter settings, the normalization, the $\log_2$ declaration, the imputation settings and the imputed matrix, and the phosphoprotein aggregation mode.
- The condition palettes of all three levels.
- The interface: the section and tab that were open, the right-hand panel, whether advanced plot mode was on and the appearance settings of each plot, and the per-page settings each panel keeps — selected levels, data sources, seeds and the rest.

What does not

- The measured values. Only filenames and, in an ordinary state, paths are recorded.
- The $\log_2$ and filtered matrices. They are recomputed from the re-read dataset and the restored filter and normalization settings, which is why a restore can take a moment on a large table. The imputed matrix, which could not be recomputed identically without its draw, is carried in the file.
- Anything belonging to the installation rather than to the analysis: the settings pages, access configuration and viewer configuration are not written into a save state and are not touched by restoring one.
- Your uploaded files. A restore reads them and never writes to them, so a state restored against the wrong data changes the analysis, not the data.

4 How it is applied

HOW IT IS COMPUTED

The file is a gzip-compressed JSON document carrying a format tag, the application version that wrote it, the time it was written and the portable flag. Restoring reads it, then checks every dataset it refers to before touching anything: if one is missing and no replacement was supplied, the whole restore is refused and the missing files are reported.
When everything needed is present, the session's datasets, annotations, uploaded metadata, condition palettes and peptide metadata are cleared, and then rebuilt from the file. The annotation for each level is recomputed from the restored dataset and its stored settings. If anything fails part-way, every one of those stores is put back exactly as it was before the restore began, and the error is reported: a failed restore never leaves half a session behind.

LIMITS

A state file larger than about 64 MB, or one that expands beyond about 256 MB, is refused, as is a replacement set beyond the configured number of files or total size. Those are the defaults; whoever runs the instance can raise them.

Failure modes

- *An empty file, or one that is not this format.* Reported as such: the state must carry the `copylot_state_v1` tag.
- *A missing dataset.* The **Missing Files** panel, as above. A portable state always lands here, for every dataset it holds.
- *A dataset that could not be restored.* Its annotation is skipped with a warning naming the package, profile and level, rather than being restored against nothing.
- *A stored imputed matrix whose sample columns no longer match the annotation.* Discarded with a warning, and the imputation is marked as not applied; run imputation again.
- *A state written by a different release.* Restored, with a warning giving both versions. Settings that release did not have take the current defaults, which is stated in a notice on every restore.

EASY TO GET WRONG

A state written before a breaking change restores the same settings, and those settings can now mean something different. The concrete case is fold-change orientation: the difference is now computed as condition 1 minus condition 2, so a volcano regenerated from an older state selects the same two conditions and comes out with every sign re-

versed against the figure that state was saved beside. Magnitudes, *p*-values and counts are unaffected — only the direction is. When a restore reports a version difference, regenerate the figures rather than trusting the ones already in the manuscript.

5 Exporting

The save state *is* the export: there is nothing else to download from this page. Keep it beside the data it refers to, since an ordinary state is only as good as the paths it records, and prefer a portable state plus the raw files when the analysis is being handed over.

Behind this page

Function	What it produces
export_state_blob	Builds the save state, with or without local paths
decode_restore_blob	Reads and validates the file
inspect_missing_files	The datasets a state refers to and cannot find
apply_restore_state	Applies the state, or rolls the session back unchanged
migrate_restore_ui_state	The disclosed migrations of older interface settings